UNIT 4

INTEGERS

4.1 INTEGERS

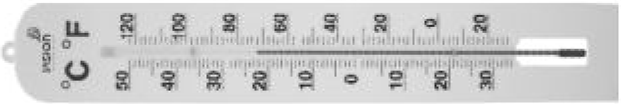
Know that the natural numbers 1, 2, 3, ..., are also called positive integers and the corresponding negative numbers –1, –2, –3, ..., are called negative integers

We are familiar with natural numbers 1, 2, 3, ... These natural numbers are also called positive integers and are used mainly to count and measure the quantities, for example:

- 5 kilograms of apples,
- 15 litres of milk etc.

Similarly there are corresponding negative numbers –1, –2, –3, ... which are called negative integers and are used mainly to measure quantities for example:

- –2 degree centigrade temperature
- –5 metres altitude
  (5 m below sea level)



> The natural numbers 1, 2, 3, ... are also called **positive integers** and the corresponding numbers –1, –2, –3, ... are called **negative integers**.

Know that '0' is an integer which is neither positive nor negative

Besides positive and negative integers, there is an integer which is neither positive nor negative that is '0'.

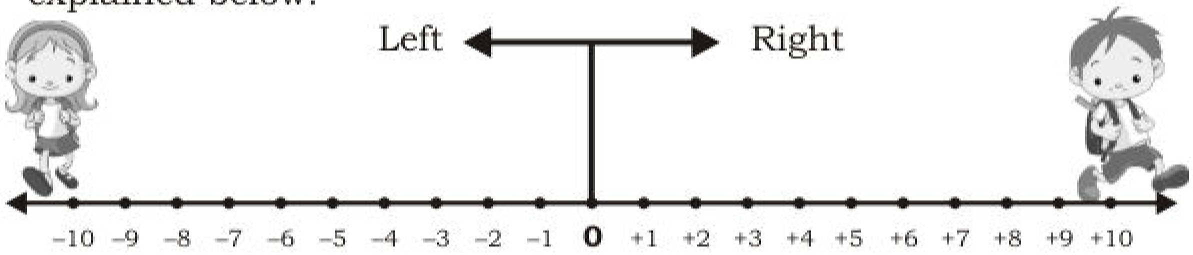
> '0' is an integer which is neither positive nor negative.

Recognize integers

All the integers are as under:
..., –3, –2, –1, 0, 1, 2, 3, 4, ...

Teacher's Note *Teacher should give some more examples of directed numbers from daily life.*

These integers are also called **directed numbers** and they are used to represent distance alongwith the direction or position, as explained below:



A boy and girl starting walking from same point 0. One goes on right covers 10 m and other goes on left covers 10 m. The starting point is represented by "0". The distance of 10 m on the right side of starting point is shown by +10 m. The distance of 10 m on the left side of starting point is shown by –10 m.

Example: Fill in the blanks.

(i) If +5 m represents, distance of 5 m towards east then distance of 10 m towards west is **–10 m**.

(ii) If –6 m represents distance of 6 m below sea level then the distance of 8 m above sea level is **+8 m**

Activity **By using integers fill in the blanks.**

(i) If –10 m represents, distance towards west then **+10 m** represents distance towards east.

(ii) If +6 million shows increase in population then ________ shows decrease in population.

(iii) If +20 m shows 20 m distance above sea level then _______ shows 50 m below sea level.

(iv) If –20 rupees represents loss then __________ represents profit in a business.

(v) If –50 m represents, distance of 50 m towards south then _________ represents 60 m towards north.

4.2 ORDERING OF INTEGERS

Represent integers on number line

We already know how to represent a natural number on a number line. Let us represent integers on a number line.

This line is called a number line.

Note: The space between any two points or numbers is always same.

Know that on the number line any number lying to the right of zero is positive

Consider a number line.

We observe that all numbers to the right of zero are positive.

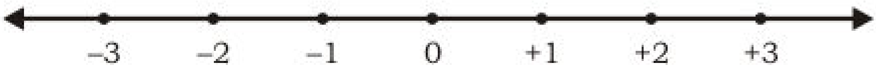
For example: +1 is at right of zero or $+1 > 0$.

+2 is at right of zero which is positive or $+2 > 0$ and so on.

Thus $\boxed{\text{Any number lying to the right of zero is positive.}}$

Know that on the number line any number lying to the left of zero is negative

Let us consider number line.



On this line we observe that all numbers to the left of zero are negative.

For example: –1 is to the left of zero. So –1 is negative or $-1 < 0$.
–2 is to the left of zero so, –2 is negative or $-2 < 0$ and so on.

Hence $\boxed{\text{On a number line, any number to left of zero is negative.}}$

Know that on the number line any number lying to the right of another number is greater

Let us again consider the number line.

We observe that any number to the right of any other number is always greater.

For example +3 is to the right of +1. So, +3 is greater than +1. i.e. $+3 > +1$

Hence

On a number line, any number to the right of another number is greater.

Know that on the number line any number lying to the left of another number is smaller

Let us consider the same number line once again.

We observe that any number to the left of any other number is smaller. For example: –2 is to the left of +1 so, –2 is smaller than +1 i.e. $-2 < +1$.

Hence

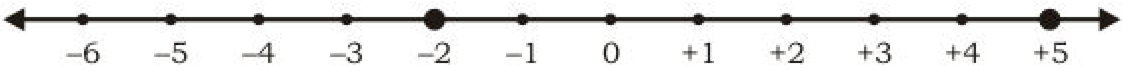
On a number line any number to the left of another number is smaller.

Example: Decide whether first number is greater or smaller using number line.

(i) +5, –2 (ii) –6, 0 (iii) +4, 0 (iv) –3, –5

Solution:

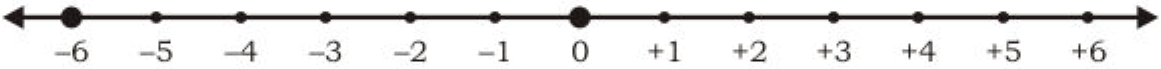
(i) Let us represent the numbers on number line.



The number +5 is at the right of –2

Therefore $+5 > -2$.

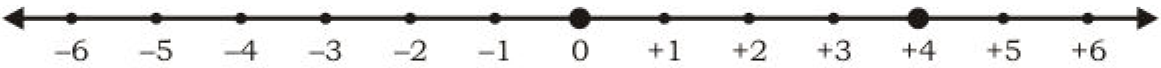
(ii) Let us represent the numbers on number line.



The number –6 is at the left of zero.

Therefore –6 < 0.

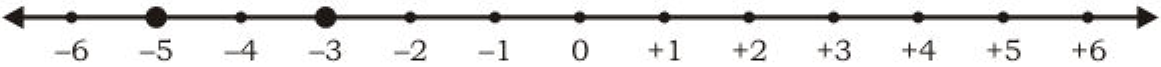
(iii) Let us represent the numbers on number line.



The number +4 is at the right of zero.

Therefore +4 > 0.

(iv) Let us represent the numbers on number line.



The number –3 is at the right of –5.

Therefore –3 > –5.

Activity 1 **Fill in the blanks from given options.**

(i) –2 is on **left** of +3. (left, right)

(ii) +5 is on ________ of –10. (left, right)

(iii) 0 is on ________ of –6 (left, right)

Activity 2 **Fill in the blanks from given options.**

(i) +5 __>__ –5 (<, >)

(ii) +7 ________ +11 (<, >)

(iii) –9 ________ –4 (<, >)

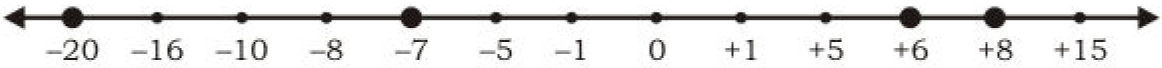
(iv) +2 ________ –20 (<, >)

Know that every positive integer is greater than a negative integer and every negative integer is less than a positive integer

Let us observe the relation between negative and positive numbers from the following examples:

(i) $+6 > -7$ (ii) $+1 > -20$ (iii) $+8 > -1$

Let us represent the number on number line.



It is observed that:

(i) $+6 > -7$ because $+6$ is on right of -7

(ii) $+1 > -20$ because $+1$ is on right of -20

(iii) $+8 > -1$ because $+8$ is on right of -1

From above examples it is clear that all positive numbers are on right of negative numbers.

Hence,

Every positive integer is greater than a negative integer.

From number line it is obvious that all negative numbers are on the left of positive numbers.

Hence,

Every negative integer is less than a positive integer.

For example: (i) $-2 < +6$ (iv) $+7 > -2$

(ii) $-3 < +1$ (v) $+20 > -50$

(iii) $-36 < +2$

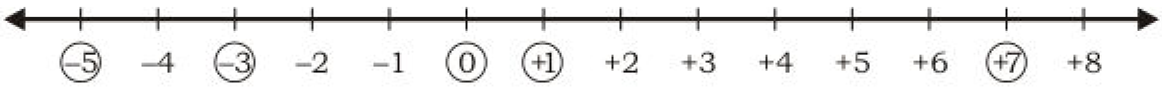
Arrange a given list of integers in ascending and descending order

Given set of integers can be arranged in ascending and descending order with the help of number line.

On a number line the right most number is the greatest and the left most number is the smallest.

Example: Arrange the following numbers –3, 0, 7, 1 –5 in ascending and descending order.

Solution: First of all we indicate the given numbers (–3, 0, 7, 1, –5) on a number line as under.



Here the greatest number is 7 and the smallest number is –5.

So, **Ascending order** (smallest to greatest) is:

–5, –3, 0, 1, 7

Descending order (greatest to smallest) is:

7, 1, 0, –3, –5

4.3 ABSOLUTE OR NUMERICAL VALUE OF AN INTEGER

Define absolute or numerical value of a number as its distance from zero on the number line and is always positive

We know that every integer except zero, on number line represents its distance from zero alongwith its direction as +5 means distance of 5 units on right of zero and –5 means distance of 5 units on the left of zero.

If we ignore direction and just consider the distance then this distance is called absolute or numerical value and we define as:

> Absolute value or numerical value of a number is its distance from zero on the number line and is always positive or zero.

Absolute value of a number is denoted by the symbol "| |"
for example $|-6| = 6$
we read as "Absolute value of –6 as 6"

Similarly
$|+7| = 7$, $|-15| = 15$ and $|0| = 0$

Arrange the absolute or numerical values of the given integers in ascending and descending order

Let us understand the ascending and descending order of absolute or numerical values of given integers with the help of example.

Example:
Write the absolute values of following numbers in ascending and descending orders.
+4, –2, +1, –3, 0, –5, +6

Solution:

Here,

$|+4| = 4$ and $|-3| = 3$
$|-2| = 2$ $|0| = 0$
$|+1| = 1$ $|-5| = 5$
$|+6| = 6$

Absolute values of the given integers are:

4, 2, 1, 3, 0, 5, 6

So, ascending order is:

0, 1, 2, 3, 4, 5, 6

And descending order is:

6, 5, 4, 3, 2, 1, 0

EXERCISE 4.1

1. **Represent the following integers on a number line.**
   (i) –2, –1, 0, +1, +2
   (ii) –3, –2, –1, 0, +1, +2, +3, +4
   (iii) –4, –3, –2, –1, 0, +1, +2
   (iv) +5, –5, –4, +3, +1, –2

2. **Decide whether first number is greater or smaller.**
   (i) +15, –6
   (ii) –8, 0
   (iii) +16, 0
   (iv) –2, –8
   (v) +7, +9
   (vi) –4, –1

3. Fill in the blanks from given options.

(i) –5 is on _______ of + 6 (left, right)
(ii) +6 is on _______ of –7 (left, right)
(iii) 0 is on ________ of –15 (left, right)
(iv) 0 is on ________ of 20 (left, right)

4. Fill in the blanks from given options.

(i) +10 _______ –20 (<, >)
(ii) –16 _______ –4 (<, >)
(iii) +25 _______ –100 (<, >)
(iv) +30 _______ +50 (<, >)
(v) –17 _______ +17 (<, >)
(vi) 0 _________ –5 (<, >)

5. Arrange the given integers in ascending and descending order.

(i) +5, –7, +1, 0, –3, –1
(ii) –3, +4, 0, –1, +2, +5
(iii) 0, –4, +4, –5, +5
(iv) –4, –1, –7, –2, –8

6. Write the absolute values of the following.

(i) –5 (ii) +20 (iii) 0 (iv) –18 (v) +50

7. Arrange the absolute values of given numbers in ascending and descending orders.

(i) –4, +1, –6, +3, 0, +5
(ii) –25, 0, +17, –10, +30, –60
(iii) –20, –10, +11, +7, 0, –4
(iv) +8, –5, +13, –9, –12, +3

8. Write True or False.

(i) Positive integer is always greater than negative integer. ( )
(ii) 0 is a positive integer. ( )
(iii) Every negative integer is less than zero. ( )
(iv) Absolute value of –5 is less than absolute of +4. ( )
(v) –5 is smaller than –10. ( )
(vi) –25 is greater than –100. ( )
(vii) Numerical value of a number can never be negative. ( )

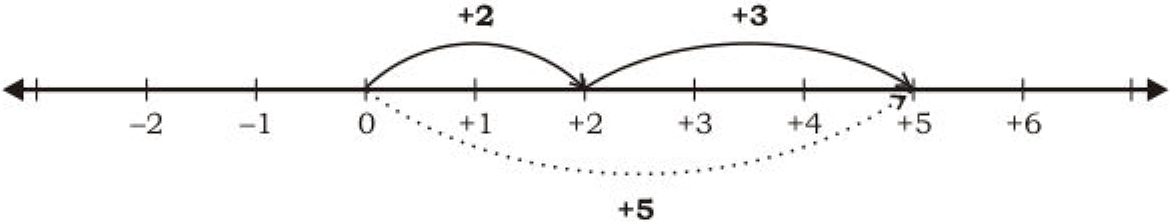
4.4 ADDITION OF INTEGERS

We have already learnt how to find sum of two or more whole numbers on the number line.

Let us revise with the help of following examples.

Example 1: Add +2 and +3 using number line.

Solution:



Here, (+2) + (+3) = +5

Explanation:
Starting from zero, first we move 2 steps on right side of zero reaching at +2.

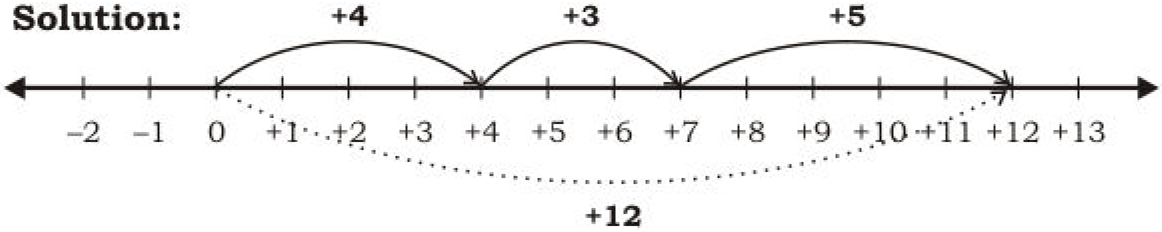
From there move 3 steps on right side reaching at +5

Thus, $\boxed{(+2) + (+3) = (+5)}$

Similarly we can add more than two whole numbers as mentioned in the following example.

Example 2: Find the sum: (+4) + (+3) + (+5) using number line.

Solution:



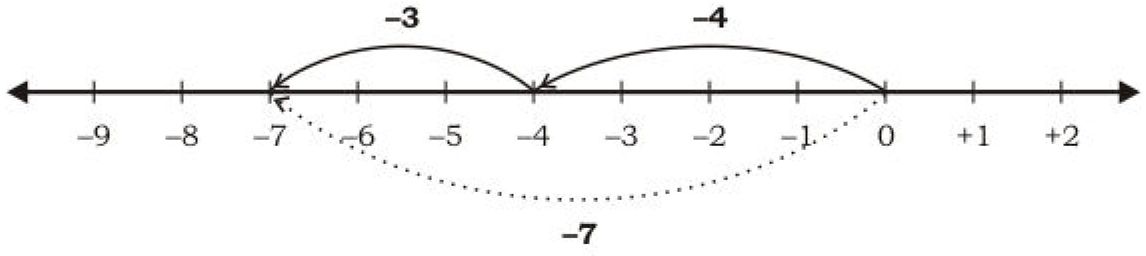
Here, (+4) + (+3) + (+5) = (+12)

Use number line to display sum of two or more given negative integers

Sum of two or more negative integers using number line is explained by the following examples.

Example 1: Find the sum of –4 and –3.

Solution:



Here, (–4) + (–3) = –7

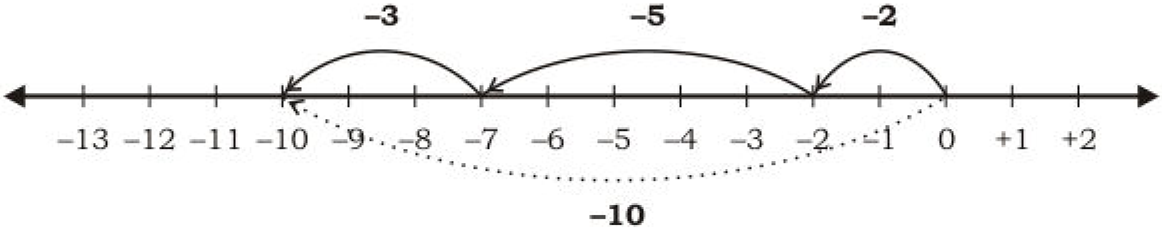
Explanation: Starting from zero, first we move 4 steps from zero on its left reaching at –4

Then from there, we move 3 more steps on left reaching at –7

Thus, $\boxed{(-4) + (-3) = -7}$

Example 2: Find the sum: (–2) + (–5) + (–3).

Solution:



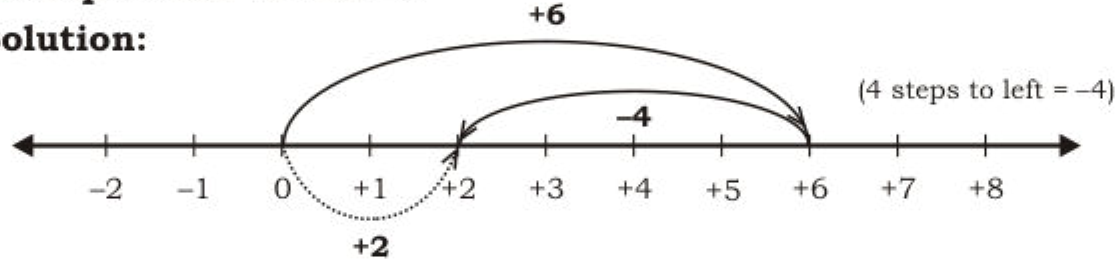
Here, (–2) + (–5) + (–3) = –10

Use number line to display difference of two given positive integers

The sum of a positive and a negative integer is explained with the help of following example.

Example: Add +6 and –4.

Solution:



Here, (+6) + (–4) = +2

Explanation:

First we move 6 steps from zero on its **right** reaching at +6.

Then from there, we move 4 steps on **left** reaching at +2.

Thus, $\boxed{(+6) + (-4) = +2}$

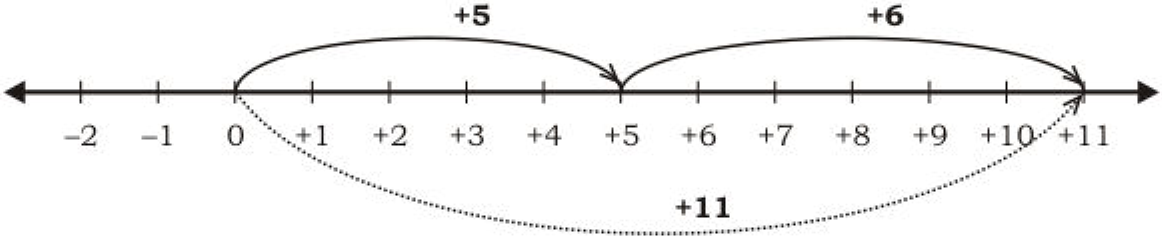
Use number line to display sum of two given integers

Sum of two given integers is explained with the help of following examples.

Example: Add:

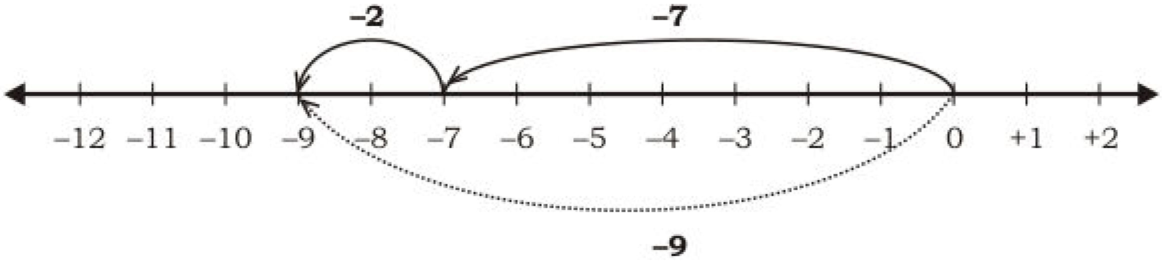
(i) +5 and +6 (ii) –7 and –2
(iii) +8 and –5 (iv) –6 and +2

Solution: (i) +5 and +6



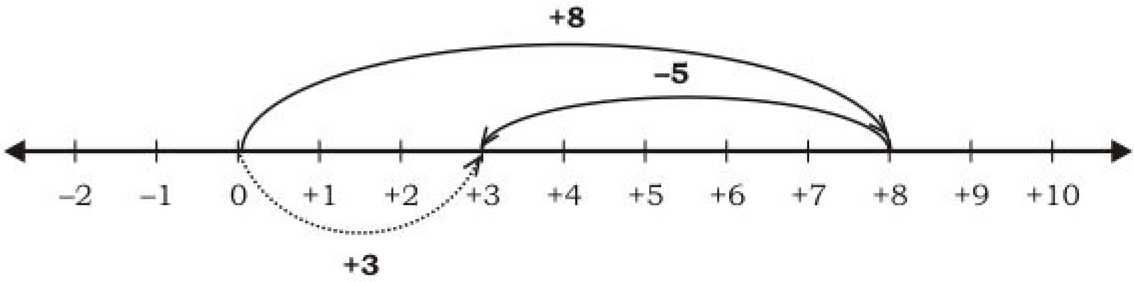
First we move 5 steps on right side of 0 reaching at +5.
Then from there, we move 6 steps on right side reaching at +11.
Here, (+5) + (+6) = +11

Solution: (ii) Add –7 and –2 means to solve (–7) + (–2)



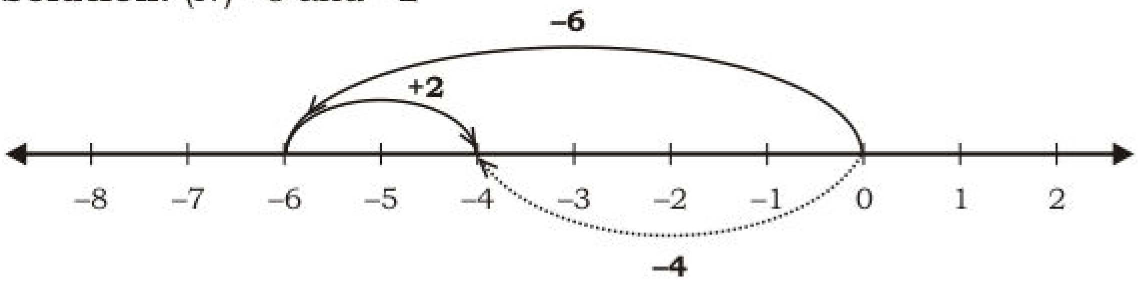
First we move 7 steps from 0 on its left reaching at –7.
Then from there, we move 2 more steps on left reaching at –9.
Here, (–7) + (–2) = –9

Solution: (iii) +8 and –5



First we move 8 steps from zero on its right reaching at (+8).
Then from there we move 5 steps on left; reaching at +3
Here, (+8) + (–5) = +3

Solution: (iv) –6 and +2



Starting from 0, first we move 6 steps from zero on its left, reaching at –6. Then from there we move 2 steps on its right; reaching at point –4.

Here, (–6) + (+2) = –4

Add two integers (with like signs)

Two integers (with like signs) are added using the following three steps.

(i) Take absolute values of given integers.

(ii) Add the absolute values.

(iii) Give the result the common sign.

Examples: Solve: (i) (+12) + (+13) (ii) (–10) + (–14)

Solution:

(i) $(+12) + (+13)$

Here, $|+12| = 12$

$|+13| = 13$

Now $(+12) + (+13)$

$= +(12 + 13)$

$= +25$

(ii) $(-10) + (-14)$

Here, $|-10| = 10$

$|-14| = 14$

Now $(-10) + (-14)$

$= -(10 + 14)$

$= -24$

Add two Integers (with unlike signs)

Two integers (with unlike signs) are added using the following three steps.

(i) Take absolute values of given integers.

(ii) Subtract the smaller absolute value from the larger.

(iii) Give the result the sign of integer with the larger absolute value.

Examples: Solve: (i) (+15) + (–10) (ii) (–20) + (+12)

Solution:

(i) $(+15) + (-10)$

Here, $|+15| = 15$

$|-10| = 10$

So, $(+15) + (-10)$

$= +(15 - 10)$ (Sign of integer wtih greater absolute value is +)

$= +5$

(ii) $(-20) + (+12)$

Here, $|-20| = 20$

$|+12| = 12$

So, $(-20) + (+12)$

$= -(20 - 12)$ (Sign of integer with greater absolute value is –)

$= -8$

Teacher's Note *Teacher should ask some easy oral questions for good practice of addition of integers.*

EXERCISE 4.2

1. Find the sum using number line.

(i) (+2) + (+7) (ii) (–5) + (–6) (iii) (+7) + (–4)

(iv) (–8) + (+2) (v) (+5) + (– 8) (vi) (+9) + (–9)

2. Find the sum using number line.

(i) (+5) + (+2) + (+3) (ii) (–5) + (–3) + (–4)

(iii) (+4) + (+5) + (+1) (iv) (–2) + (–6) + (–5)

3. Solve the following.

(i) (+10) + (+20) (ii) (–15) + (–25) (iii) (–20) + (–7)

(iv) (+5) + (–14) (v) (–14) + (+8) (vi) (–30) + (+30)

4.5 SUBTRACTION OF INTEGERS

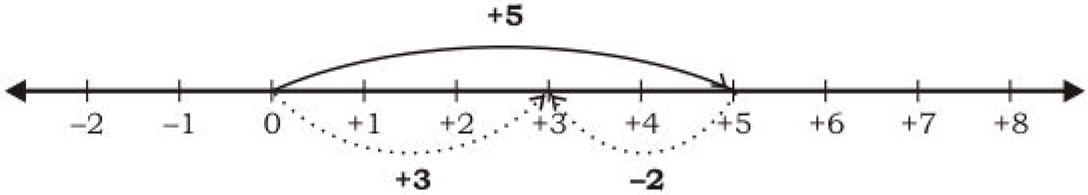
Recognize subtraction as the inverse process of addition

Subtraction is the reverse process of addition because subtracting a subtrahend from a given number means adding the subtrahend with reverse direction in given number.

Let us understand with the following example.

Example 1: Subtract +2 from +5.

Solution: First of all we change the sign of +2 and get –2 then add –2 in +5 on number line.



So, (+5) – (+2)

= (+5) + (–2)

= +3

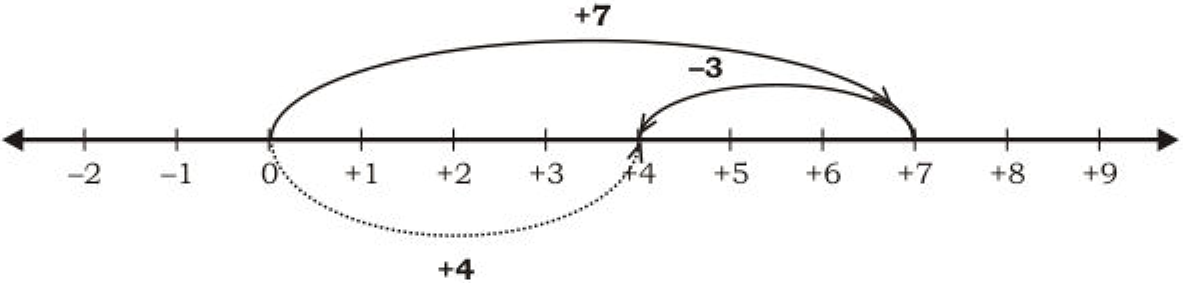
Hence in order to subtract +2 from +5 we have added +2 with reverse direction (i.e. –2) in +5.

So, we say subtraction is the reverse process of addition.

Example 2: Subtract +3 from +7 using number line.

Solution: To subtract +3 from +7 means;

to solve: (+7) – (+3)



Here, (+7) – (+3) = +4

Subtract one integer from the other by changing the sign of the integer being subtracted and adding according to the rules for addition of integers

It is clear from previous example that if we subtract one integer from other then we have to follow the following rule.

Subtract one integer from the other by changing the sign of the integer being subtracted and adding according to the rules for addition of integers.

Example 1: Subtract: (i) +2 from +6 (ii) –3 from +8

Solution:

(i) +2 from +6

Subtract +2 from +6 means to find difference between (+6) and (+2)

Difference	=	(+6) – (+2)	
	=	(+6) + (–2)	(changing sign and adding)
	=	+ (6 – 2)	$\because$ $\lvert +6 \rvert = 6$
	=	+ 4	$\lvert -2 \rvert = 2$

(ii) –3 from +8

Subtract –3 from +8 means to find difference between (+8) and (–3)

Difference	=	(+8) – (–3)	
	=	(+8) + (+3)	(changing sign and adding)
	=	+ (8 + 3)	$\because$ $\lvert +8 \rvert = 8$
	=	+ 11	$\lvert +3 \rvert = 3$

Note: We can solve (+6) + (+2) by adding inversely as under:
(+6) + (+2) = (+6) – (–2) = (+6) + (+2) = +(6+2) = +8

Example 2: Solve: (i) (–8) – (–5) (ii) (+7) – (+10)

Solution:

(i) (–8) – (–5)

= (–8) – (–5)
= (–8) + (+5) (changing sign and adding)
= – (8 – 5) $\because$ |–8| = 8
= – 3 |+5| = 5

(ii) (+7) – (+10)

= (+7) – (+10)
= (+7) + (–10) (changing sign and adding)
= – (10 – 7) $\because$ |–10| = 10
= – 3 |+7| = 7

EXERCISE 4.3

1. Find the difference using number line.

(i) (+6) – (+4) (ii) (–8) – (–3) (iii) (–9) – (–3)
(iv) (–9) – (+5) (v) (+9) – (–2) (vi) (– 7) – (+4)

2. Subtract.

(i) +6 from –10 (ii) –10 from –20 (iii) –8 from –7
(iv) 15 from +20 (v) –6 from + 14 (vi) –9 from –6

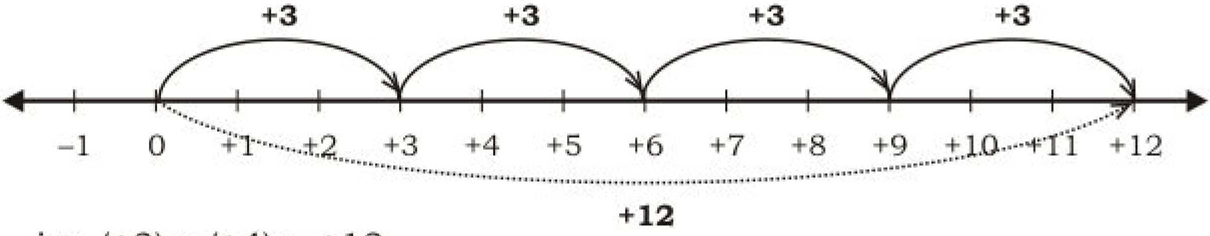
3. Solve the following.

(i) (+25) – (–15) (ii) (–30) – (–25) (iii) (–11) – (+5)
(iv) (+40) – (+30) (v) (–16) – (–18) (vi) (+15) – (–20)
(vii) (–29) – (+17) (viii) (–20) + (–30) (ix) (–27) + (+17)

4.6 MULTIPLICATION OF INTEGERS

We know that multiplication is the simple form of repeated addition for example multiplying +3 by +4 means adding +3 four times i.e. $(+3) \times (+4) = (+3) + (+3) + (+3) + 3$

Representation on number line



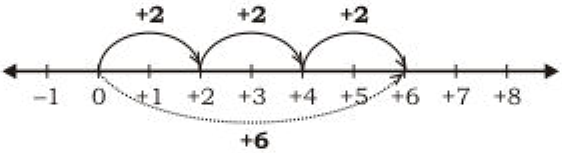
i.e. $(+3) \times (+4) = +12$

Recognize that the product of two integers of like signs is a positive integer

Consider the following examples of multiplication using number line.

(i) $(+2) \times (+3)$

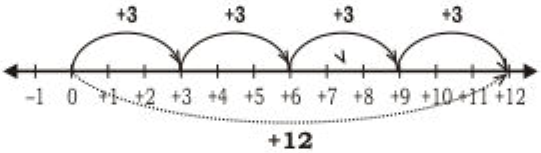
Multiplying +2 with +3 means adding +2 three times in same direction of +2.
i.e.



Here $(+2) \times (+3) = +6$

(ii) $(-3) \times (-4)$

Multiplying –3 with –4 means adding –3 four times in opposite direction of –3.



$(-3) \times (-4) = (+12)$

From these two examples we observe that.

> **Rule 1:**
> The product of two integers of like signs is a positive integer.

For example:

(i) $(+3) \times (+4) = +12$ (ii) $(-3) \times (-6) = +18$

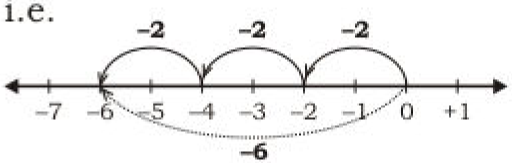
Recognize that the product of two integers of unlike signs is a negative integer

Consider the following examples of multiplication using number line.

(i) $(-2) \times (+3)$

Multiplying –2 with +3 means adding –2 three times in same direction of –2.

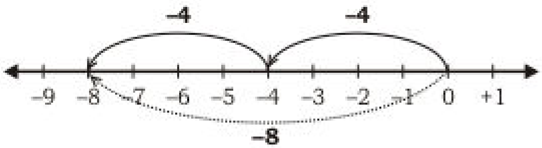
i.e.



So, $(-2) \times (+3) = -6$

(ii) $(+4) \times (-2)$

Multiplying +4 with –2 means adding +4 two times in opposite direction of +4.



So, $(+4) \times (-2) = -8$

From these two examples we observe that.

> **Rule 2:**
> The product of two integers of unlike signs is a negative integer.

For example:

(i) $(-2) \times (+5) = -10$ (ii) $(+3) \times (-5) = -15$

In multiplication of two integers we multiply the absolute values of integers and apply the rules of multiplication of integers.

Examples: Find the product.

(i) $(+15) \times (-23)$ (ii) $(-25) \times (-14)$

Solution:

(i) $(+15) \times (-23)$

$= -(15 \times 23)$ (Applying Rule 2)

$= -345$

$$\begin{array}{r} +15 \\ \times\ -23 \\ \hline 45 \\ 30\text{x} \\ \hline -345 \\ \hline \end{array}$$

(ii) $(-25) \times (-14)$

$= +(25 \times 14)$ (Applying Rule 1)

$= +350$

$$\begin{array}{r} -25 \\ \times\ -14 \\ \hline 100 \\ 25\text{x} \\ \hline +350 \\ \hline \end{array}$$

Note: Product of any integer and zero is always zero.

4.7 DIVISION OF INTEGERS

Recognize that division is the inverse process of multiplication

We know that the operation of division is the inverse process of multiplication for example dividing +20 by +5 implies that to find a number which when multiplied by +5 gives +20 as product.

i.e. $\frac{20}{5} = \boxed{} \Rightarrow 20 = \boxed{} \times 5$

Same number

Guess it?

This number is 4.

because $\frac{20}{5} = \boxed{4} \Rightarrow 20 = \boxed{4} \times 5$

Recognize that on dividing one integer by another if both the integers have like signs the quotient is positive

Remember that

> Dividing a number with any other number is in fact the product of given number with reciprocal of the divisor.

For example:

(i) $20 \div (+2) = 20 \times (+\frac{1}{2})$

$= \frac{20}{2}$

$= 5$

So, $\boxed{20 \div (+2) = +5}$

(ii) $(-18) \div (-3) = (-18) \times (-\frac{1}{3})$

$= +\frac{18}{3}$

$= +6$

So, $\boxed{(-18) \div (-3) = +6}$

From these two examples we observe that:

> **Rule 1:** For integers of like signs, the quotient is positive.

For example:

(i) $(+10) \div (+2) = +5$

(ii) $(-20) \div (-5) = +4$

Recognize that on dividing one integer by another if both the integers have unlike signs the quotient is negative

Let us consider the following examples of division.

(i) $(+8) \div (-2)$

$= (+8) \times (-\frac{1}{2})$

$= -\frac{8}{2}$

$= -4$

So, $\boxed{(+8) \div (-2) = -4}$

(ii) $(-15) \div (+3)$

$= (-15) \times (+\frac{1}{3})$

$= -\frac{15}{3}$

$= -5$

So, $\boxed{(-15) \div (+3) = -5}$

From these two examples we observe that:

> **Rule 2:** For integers of unlike signs, the quotient is negative.

For example:

(i) $(+6) \div (-2) = -3$ (ii) $(-8) \div (+4) = -2$

In division of one integer by another, we just perform the division of their absolute values and apply the rules of division.

Examples: Find the quotient.

(i) $(-36) \div (-4)$ (ii) $(+54) \div (-2)$

Solution:

(i) $(-36) \div (-4)$

$= (-36) \div (-4)$

$= +(36 \div 4)$ (Applying Rule 1)

$= +9$

$$\begin{array}{r} 4\overline{)\,36\,}(9 \\ -36 \\ \hline 0 \end{array}$$

(ii) $(+54) \div (-2)$

$= (+54) \div (-2)$

$= -(54 \div 2)$ (Applying Rule 2)

$= -27$

$$\begin{array}{r} 2\overline{)\,54\,}(27 \\ -4 \\ \hline 14 \\ -14 \\ \hline 0 \end{array}$$

Teacher's Note *At this level, teacher should only consider division of those integers which are perfectly divisible.*

Know that division of an integer by '0' is not possible

Let us consider division of any integer by 0 for example.

Example: Divide +5 by 0.

Solution:

Dividing +5 by 0 implies to find a number by which when 0 is multiplied the product should be 5

$$\begin{array}{r|l} 0\overline{)\;5\;}(1 & \\ -0 & \\ \hline 5 & \end{array} \quad , \quad \begin{array}{r|l} 0\overline{)\;5\;}(2 & \\ -0 & \\ \hline 5 & \end{array} \quad , \;\_\_\_\_ \;,\; \_\_\_\_ \;,\; \_\_\_\_$$

This process of division will continue and never finish.

It is impossible as such number does not exist.
Hence division of any integer by 0 is not possible.

EXERCISE 4.4

1. Find the product.

(i) $(+15) \times (-4)$ (ii) $(+20) \times (+17)$ (iii) $(-16) \times (-25)$

(iv) $(-36) \times (+12)$ (v) $(+35) \times (-14)$ (vi) $(-43) \times (-16)$

2. Find the quotient.

(i) $(+25) \div (-5)$ (ii) $(-35) \div (-7)$ (iii) $(+100) \div (+4)$

(iv) $(-252) \div (+4)$ (v) $(-234) \div (+3)$ (vi) $(-126) \div (-6)$

3. Write true or false.

(i) $+5 \div 0 = 0$ ( ) (ii) $0 \div +6 = 0$ ( )

(iii) $+9 \times 0 = +9$ ( ) (iv) $+6 \times 0 = 0$ ( )

(v) $(-10) \div (-10) = -1$ ( ) (vi) $(-20) \times (-1) = +20$ ( )

(vii) $(+5) \div (-5) = 1$ ( ) (viii) $(-18) \div (-3) = (+6)$ ( )

REVIEW EXERCISE 4

1. Write all numbers between –5 and + 5.

2. Represent the following integers on a number line.
   (i) –5, –2, 0, +3, +5 (ii) –6, –3, –1, +1, +4, +6

3. Write the absolute values of:
   (i) –3, +4, –4, +6, 0 (ii) +2, –7, +1, 0, +4, –2

4. Find the sum using number line.
   (i) (+6) + (–9) (ii) (–9) + (–6)
   (iii) (+9) + (+7) + (–5) (iv) (–1) + (–2) + (3)

5. Find the difference using number line.
   (i) (+5) – (+3) (ii) (–7) – (–2) (iii) (+4) – (–3)

6. Solve the following:
   (i) (+6) + (–10) (ii) (–10) + (–4)
   (iii) (–8) + (+5) (iv) (+20) – (–10)
   (v) (–20) – (–15) (vi) (+40) – (+25)

7. Solve the following:
   (i) (+10) × (–5) (ii) (–6) × (–9)
   (iii) (+20) × (+6) (iv) (–15) × (+3)
   (v) (+6) × 0 (vi) (–10) × (–8)

8. Find the quotient.
   (i) (+20) ÷ (–4) (ii) (–15) ÷ (–3)
   (iii) (–36) ÷ (+6) (iv) (+40) ÷ (+10)

SUMMARY

- +1, +2, +3, ... are called positive integers.
- –1, –2, –3, ... are called negative integers.
- 0 is neither positive nor negative.
- Any number to the right of another number is greater.
- Any number to the left of another number is smaller.
- Every positive integer is greater than a negative number.
- Every negative integer is less than a positive integer.
- Absolute value of a number is always non-negative.
- Sum of two positive integers is always positive integer.
- Sum of two negative integers is always negative integer.
- Product of two integers of like sign is always positive integer.
- Product of two integers of unlike signs is always negative integer.
- Product of zero and any other integer is always zero.
- For integers of like signs, the quotient is positive.
- For integers of unlike signs, the quotient is negative.
- Division by zero is not possible.